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Cosmology from the CMB with advanced sampling techniques

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Sampling the Cosmic Microwave Background using Advanced Sampling Techniques

CMB (Cosmic Microwave Background) photons are prolific in the Universe. But what actually is the CMB? It is a set of photons whose origins date back to the early Universe and can today be observed everywhere, with an average temperature, $T_\gamma = 2.7255 \pm 0.0006\text{K}$.

Despite the photons being more common than the atoms surrounding us, the discovery of the CMB was a fortunate accident by scientists Arno Penzias and Robert Wilson in 1964 [1]. Having built a radiometer (basically a big device to measure the power of light around you), they observed 'noise' in the which they could not remove (even when they scared away some nearby pigeons). This was later determined to be the CMB.

CMB measurements have since developed into a multimillion-pound sector of research, with the most recent satellite, 'Planck', costing approximately £600m and with this increase in spending, we have been able to significantly develop our understanding.

One of the reasons for its relatively late discovery is the low energy of the photons - the only real tangible impact they have on modern-day life is a small amount of static on your TV [2]. However, this is misleading as we know they are so much more important than they appear.

The CMB is not quite isotropic; there are minor fluctuations (of order 10^{-5} of the mean temperature), the causes of which can be attributed to the early Universe. Measurements of these fluctuations can give information about the parameters which describe the Universe (such as the density of matter and dark matter), and so these fluctuations need to be measured with great precision to obtain valid predictions.

The problem is that the anisotropies are so small - requiring high-resolution measurements, and are at such low temperatures, they are susceptible to external noise. We do, however, have a trick to solving this - the power of Bayesian statistics!

Bayesian statistics allow the inference of a set of parameters from a data set. What this means is that we can use statistics to infer the true CMB. This inference comes from sampling a 'posterior' function, but this was not the novelty of the work undertaken. In this project, a model was built to sample this posterior function with advanced sampling techniques.

Previous attempts at sampling this posterior are very computationally expensive, making the inference of this power spectrum unfeasible. The aforementioned 'advanced sampling techniques' consist of optimised functions and automatic differentiation provided by 'Tensorflow' (an opensource software library used for Python and C++). Using these, the sampling of this 'posterior' function can be drastically increased and indeed were. The model sampled four independent chains, each with 2400 samples - with a total sampling time of 138 hours (taking approximately 51.8s per step). The mean acceptance rate of the chains was 0.985, indicating optimised sampling parameters and thus efficient sampling in the parameter space. The final parameters and their error distribution were then compared to that predicted by the most widely accepted cosmological model - the ΛCDM model. These values seemed to match demonstrating a successful model.

The obstacle with the work undertaken was that for full analysis of the best data of the sky (from the Planck satellite), 149GB of RAM are needed. The machine that run the code had just 25.5GB of RAM and so less sampling parameters could be used than that of the full sky - something which could be implemented as an extension in the future!

References:

- [1] - M. Bucher, Status of CMB observations in 2015, 16/06/2016, Pg 2-5.
- [2] - B. Ryden, Introduction to Cosmology, 01/13/2006, Pg 179-184.

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Abstract

The cosmic microwave background (CMB) has been a prime source of cosmological information over the past 50 years as a result of its anisotropies links to events in the early universe. Satellites observe and record these anisotropies; however, the pixel data is noisy and true values are uncertain. A model was built to sample in harmonic space of the CMB, using a No-U-Turn Sampler (NUTS) to infer its angular temperature power spectrum. This was conducted using a map pixelised at $N_{side} = 128$ for multiple numbers $2 \leq \ell < 60$. This model was run on the Google Colab network with 25.5GB of memory and a dual-core processor. Four independent chains were obtained, each with 2400 samples - with a total sampling time of 138 hours, with an average of 54.3s per effective step (with the burn-in stage of the sampling included). The mean acceptance rate of the chains was 0.985, indicating optimised sampling parameters and thus efficient sampling in the parameter space. This inferred power spectrum and its error distribution were then compared to the power spectrum predicted by the Λ CDM model and they seemed to match to a good order. The next step would be to apply this model to a more powerful machine and then estimate the cosmological parameters from the inferred power spectrum (Chapter 5).

Contents

1	Introduction	3
2	Background Material	5
2.1	Cosmology:	5
2.2	Bayesian Analysis	7
2.3	Sampling Algorithms	9
2.4	Error Distributions	10
3	Method	12
3.1	Sampling	12
3.2	Tensorflow	13
3.3	Convergence	13
4	Results and Discussion	15
4.1	Convergence	15
4.2	Timing	17
4.3	Power Spectrum	18
5	Future outlook:	19
6	Conclusion	20
A	CMB Temperature Map Measurements	26
B	Leapfrog Method	28

List of Figures

1.1	The increase in resolution of measurement of different CMB satellites. The COBE, WMAP and Planck satellites were launched in 2008, 2013 and 2018 respectively. These maps are in the Molleview projection in the RING scheme [44]. Adapted from [29].	3
1.2	The TT power spectrum per logarithmic interval for the Planck data (red points with errorbars). This plot is overlayed with the power spectrum predicted by the Λ CDM cosmological model (blue). It has a dashed line indicating where their hybrid approach is split. Adapted from [27].	4
2.1	Comparison of the efficiency with which the NUTS and HMC sampling algorithms investigate a given sampling space. Adapted from [59].	10
4.1	Sampling of the log posterior. This has the structure of an initial burn-in stage, followed by a consistent sampling demonstrating convergence.	15
4.2	Sampling of the log posterior cut to the first 200 steps. This has the structure of an initial burn-in stage, followed by a consistent sampling demonstrating convergence and lead to a cut-off burn-in value to be chosen as 42.	16
4.3	Histograms for the Gelman-Rubin statistic for each of the sampling parameters used in the model - C_ℓ (left) and a_{lm} (right). The number of bins used in each histogram was optimised to be 7 and 43 respectively.	16
4.4	Extrapolated time taken to sample as a function of the two model parameters, by pre-loading the spherical harmonic matrix.	17
4.5	Power spectrum yielded by the Planck team through Gibbs sampling, compared to that of the Λ CDM model and the inferred power spectrum. This demonstrated an alignment between the spectra highlighting the efficacy of HMC and NUT samplers in this context. . . .	18
A.1	The brightness of sources of radiation in the milky way as a function of frequency. There is a minimum in the interfering signals between 70-80GHz where the CMB signal is greater than the total galactic signal (Sum fg). The nine different frequency bands that Planck simultaneously records in are outlined in grey [46].	26

Chapter 1

Introduction

In 1964, astronomers Robert Wilson and Arno Penzias pointed their Radiometer to the sky and observed a constant background signal which they could not eliminate. They had discovered relic radiation with an origin dating back to the early Universe [8]. Little did they know, this was the Cosmic Microwave Background (CMB) - an isotropic BlackBody spectrum with temperature, $2.72548 \pm 0.00057\text{K}$ [4]. This radiation has since become the most accurately recorded Blackbody in human history and gives cosmological information such as the conditions and origins of the early Universe [52].

The CMB was first detected to be anisotropic by the COBE satellite in 1992, and these fluctuations have been recorded with increasing resolution in the decades since, Figure 1.1. They provide cosmological information about the composition of the Universe, but to make accurate predictions need to be recorded with incredible precision. However, these fluctuations are of order of magnitude $\frac{\delta T}{T} = 10^{-5}$ (where T is the mean temperature of the sky and δT is the difference from this mean), which make them very sensitive to noise. In addition to this, there are two other obstacles when observing the CMB - other galactic sources of microwave radiation and obstacles in the sky. To obtain the underlying CMB map from measurements, a process called 'component separation' is completed on the data received from the sky (refer to Appendix A).

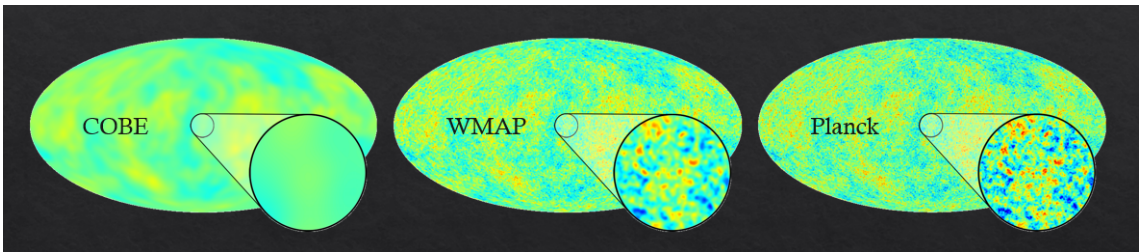


Figure 1.1: The increase in resolution of measurement of different CMB satellites. The COBE, WMAP and Planck satellites were launched in 2008, 2013 and 2018 respectively. These maps are in the Mollweide projection in the RING scheme [44]. Adapted from [29].

The most recent satellite to observe the CMB is Planck (launched in 2009), which carried an array of 74 detectors sensitive to a range of frequencies between approximately 25 and 1000 GHz. By recording the sky in a spectrum of frequencies,

the galactic foreground can be determined more accurately to return the underlying CMB map. These detectors continuously and simultaneously observe the sky temperature with an angular resolution varying between approximately 30 and 5 arcminutes (with the greater resolution at the highest frequencies and the lesser at the lower) [27]. This 'component separation' process also uses masks that remove regions of the sky that are obstructed.

Once the component separation of the sky measurements has been conducted (refer to Appendix A), the CMB map can be analysed. For Planck, this map is pixelised at $N_{side} = 2048$ - where the number of pixels, $N_{pix} = 12N_{side}^2$. Analysis of the maps is typically conducted in harmonic space, on a small section of the sky - as to make the approximation of neglecting curvature sufficient [23]. The variance of the harmonic coefficients of the spherical harmonics in this space, can be used to form an angular temperature power spectrum, C_ℓ . This power spectrum's structure originates from multiple cosmological phenomena outlined in Chapter 2.1 and is conventionally plotted as the power per logarithmic interval, $D_\ell = \frac{\ell}{2\pi}(\ell + 1)C_\ell$, Figure 1.2. It can be used as a method to predict cosmological parameters that describe the Universe by comparing power spectra that arise from different parameter values (Chapter 5). It is thus crucial that the CMB is measured accurately so that these parameters can be predicted accurately.

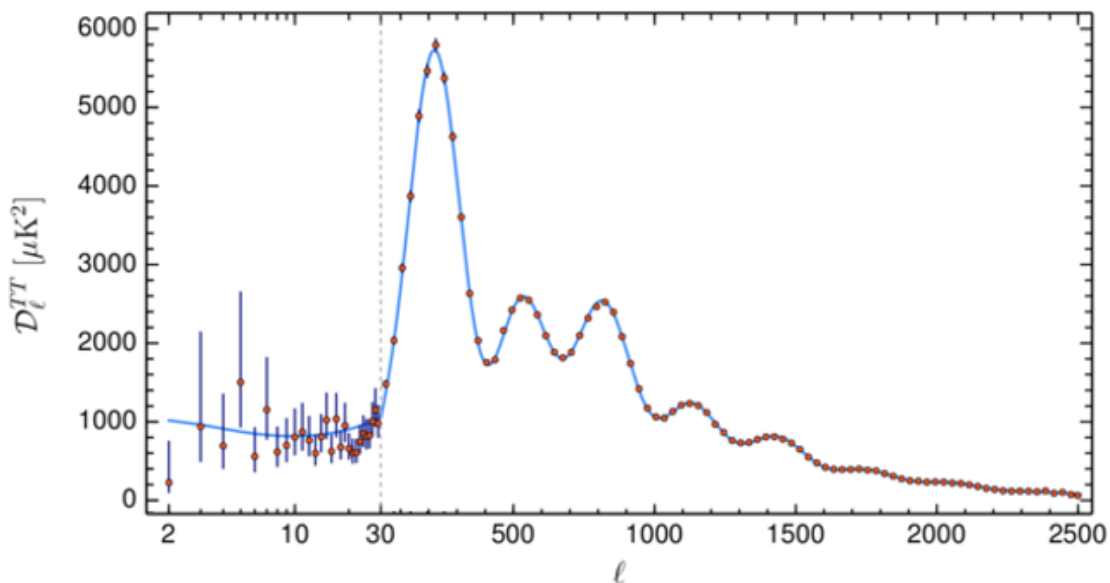


Figure 1.2: The TT power spectrum per logarithmic interval for the Planck data (red points with errorbars). This plot is overlayed with the power spectrum predicted by the Λ CDM cosmological model (blue). It has a dashed line indicating where their hybrid approach is split. Adapted from [27].

Bayesian techniques allow the inference of parameters from a data set. This is achieved through the creation of a 'posterior' function which is sampled (refer to Chapter 2.3). Sampling is used to form the pdf of a function which is not known analytically [28]. The parameters form the domain of this pdf and can be inferred from the pdf. For the inference of the power spectrum, the parameters are C_ℓ .

Chapter 2

Background Material

2.1 Cosmology:

The Hot Big Bang model of the Universe states that in the beginning the Universe formed from a singularity for which, until a time of 10^{-43} s after the Big Bang, the four fundamental forces combined to form one fundamental force - the Planck epoch [48]. This early Universe was extremely small and dense and therefore also very hot. After this the forces became what they are today, and subatomic particles formed, meaning the Universe consisted of electrons, baryons and dark matter, where the electrons and protons formed a plasma - attracted by the electromagnetic force. Furthermore, photons were coupled to the protons by Thomson scattering together, forming a photon baryon fluid. The proximity of the species meant that their wavefunctions overlapped - causing quantum properties to become apparent on the macroscopic scale. From the uncertainty principle, density anisotropies arose across the Universe. The baryons of the photon-baryon fluid were gravitationally attracted to the denser regions of the Universe. The photons opposed this pull, repelled by the photon radiation pressure - creating acoustic oscillations called baryon acoustic oscillations (BAO). These photons had wavelengths that increased in size as the Universe expanded with maximum wavelength given by the maximum distance a photon could have travelled (the maximum distance for causal connection).

Following this, at approximately 10^{-37} s after the Big Bang, the inflationary model states there was a period of rapid expansion of the Universe and therefore an expansion in the size of these density fluctuations [41]. Succeeding this, the Universe continued to expand until the "recombination" epoch. Here, the Universe had expanded such that its temperature was sufficiently low that protons could capture electrons. This meant Hydrogen atoms could form and decouple photons, releasing them with temperatures proportional to their energies - which were greater in higher density regions of the photon-baryon fluid and were also greater in regions with a lower gravitational potential (since the photons lose less energy to climb out of it) [19]. Here we are referring specifically to 'Hydrogen recombination' and are making the approximation to disregard 'Helium recombination' [52]. During this recombination time, hotter, diffusing photons tended towards colder regions in space, damping the magnitude of the photons anisotropies - diffusion damping.

The 'last scattering' epoch is the time at which a photon underwent its last scat-

tering from an electron. The probability of a photon being scattered by an electron is small once the rate of expansion of the Universe is greater than the scattering rate - therefore the 'last scattering' epoch is very close to the recombination' epoch [56].

This is the physical origin of the CMB. The structure of its anisotropies are principally determined by two stages in its formation - the the Baryon acoustic oscillations and the diffusion damping. When measured today, there is also a secondary effect on the anisotropies caused by interactions since the 'last scattering' with hot gas or gravitational potentials.

Once component separation has concluded for a satellites pixelised CMB map (refer to Appendix A), it is smoothed with a Gaussian beam. This CMB data $\mathbf{d}$, can be deconstructed into the sum of the underlying signal from the CMB $\mathbf{s}$ and some corresponding noise $\mathbf{n}$:

$$\mathbf{d} = \mathbf{s} + \mathbf{n} \quad (2.1)$$

Since the photons in the CMB are a product of the BAO, they are acoustic and so $\mathbf{t}$ can be decomposed into harmonic space using spherical harmonics [23]:

$$t(\theta, \phi) = \sum_{l=2}^{l_{max}} \sum_{m=-l}^l a_{lm} Y_{lm}(\theta, \phi) \quad (2.2)$$

where,

$$Y_{lm}(\theta, \phi) = \left(\frac{(2\ell+1)(\ell-m)!}{4\pi(\ell+m)!} \right)^{0.5} e^{im\phi} P_l^m(\cos \theta) \quad (2.3)$$

(where θ, ϕ , are the polar and azimuthal angles respectively, t is the temperature recorded at a pixel with coordinates (θ, ϕ) , $Y_{lm}(\theta, \phi)$ are the spherical harmonics, a_{lm} are the spherical harmonic coefficients which since the sky is real, $a_{lm} = a_{l,-m}^*$ and $P_l^m(\cos \theta)$ are the Legendre polynomials) [10, 51]. From this, (2.1) can further be decomposed as:

$$\mathbf{d} = \mathbf{B} \mathbf{Y} \mathbf{a} + \mathbf{n} \quad (2.4)$$

(where $\mathbf{B}$ is beam smoothing matrix and $\mathbf{Y}$ and $\mathbf{a}$ are the matrix and vector of the spherical harmonics and spherical harmonic coefficients respectively). $\mathbf{n}$, can be assumed to be Gaussian [2]. As well as this, we further assume the components of its components are uncorrelated and have uniform variance σ_{pixel}^2 , where $\langle n_i n_j \rangle = \sigma_{pixel}^2 \delta_{ij}$ [24].

Following the inflationary paradigm, the CMB is described by Gaussian statistics because the large angle anisotropy is a superposition of many independant fluctuations and so distribution tends to a normal [41,49]. For an isotropic, Gaussian CMB sky, the covariance matrix, C_{lm} of the spherical harmonic coefficients is diagonal (since the ℓ modes are independent). Thus each C_ℓ mode can be described using the equation:

$$C_{lm} C_{l'm'} = \langle a_{lm} a_{l'm'}^* \rangle = C_\ell \delta_{ll'} \delta_{mm'} \quad (2.5)$$

(where the C_ℓ coefficients denote the power spectrum and are the variance of the spherical harmonic coefficients, a_{lm}) [2]. For the C_ℓ modes, a gaussian distribution is a good approximation for high ℓ because of the central limit theorem and the number of 'm' modes that are summed over ($2\ell + 1$ modes). However, at low ℓ , this gaussianity tends to break down due to the opposite problem - not enough 'm' modes. This inherent error in the spectrum, the 'cosmic variance' ΔC_ℓ , is something which cannot be overcome and is described by:

$$\Delta C_\ell = \left(\frac{2}{2\ell + 1} \right)^{0.5} C_\ell \quad (2.6)$$

There are however complications with using these scalings. The first all which is that all sources of noise increases the errors of the distribution. If the noise is also Gaussian, it can also be described using (2.6) [57]. This is the source as to why the Planck satellite has errors which increase at the resolution scale. Recall, during 'component separation', a mask removes a portion of the sky $1 - f_{sky}$ to deal with obstructions in the sky measurements. This masking further increases these errors by a factor $f_{sky}^{-1/2}$ and this resulting variance is called 'sample variance' [23, 58].

The CMB temperature power spectrum has structure dating back to its origin. This affects $\{C_\ell\}$ such the modes tend to be affected by different origins.

For the $\ell = 0$ monopole mode, the spherical harmonic term Y is reduced to 1. This is thus the mean temperature across the sky, recorded as $T_\gamma = 2.7255 \pm 0.0006 K$ [2]. This was not considered for this project and so was removed from the map's used.

The $\ell = 1$ dipole mode, is affected by our motion relative to the rest of space - incident photons are blue-shifted and those travelling away will be red-shifted. This is a useful piece of information however, we cannot measure this mode accurately because we cannot determine whether the average temperature which is measured locally is the same as that across the Universe [10]. This is therefore removed from the CMB map for analysis.

The Sachs-Wolfe (SW) effect, is an effect on the CMB photons in which they are gravitationally red-shifted. It's primary effect is on low ℓ modes and there is two different types. Non-integrated SW occurs at the surface of last scattering due to a difference in the ratio between the energy density and the matter. Integrated SW occurs between the last scattering and us.

The Λ CDM model is the most accepted cosmological model for the Universe. It parameterises the Big Bang cosmological model for which the Universe consists of matter, dark matter and dark energy. It therefore predicts a CMB TT power spectrum for the Universe which can be compared to that inferred by the project (Chapter 2.2).

2.2 Bayesian Analysis

Bayesian statistical methods model a system and update parameters when new data is collected. Bayesian inference uses Bayes formula to infer a set of parameters.

ters from a data set recorded from the system. From Bayes theorem, a posterior, $P(\text{variables}|\text{data})$ is obtained by updating a prior, $P(\text{variables})$ using a likelihood function, $P(\text{data}|\text{variables})$:

$$P(\text{variables}|\text{data}) \propto P(\text{variables}) \prod P(\text{data}|\text{variables}) \quad (2.7)$$

(where the terms are probability densities for continuous parameters) [5][7]. For this project, we aim to sample a conditional probability distribution of the power spectrum given $\mathbf{d}$, $P(\{C_\ell\}|\mathbf{d})$. This is achieved by sampling from a joint distribution of $\{C_\ell\}$ and $\mathbf{a}$, $P(\{C_\ell\}, \mathbf{a} | \mathbf{d})$, and then marginalising over $\mathbf{a}$. This joint density can be derived using Bayes theorem, (2.7):

$$P(\{C_\ell\}, \mathbf{a}|\mathbf{d}) \propto P(\mathbf{d}|\mathbf{a})P(\mathbf{a}|\{C_\ell\})P(\{C_\ell\}) \quad (2.8)$$

(where $P(\mathbf{d}|\mathbf{a})$ and $P(\mathbf{a}|C_\ell)$ are likelihood functions and $P(C_\ell)$ is the Prior).

As for the components of (2.7), the Prior term is chosen to be unity such that the maximum of the posterior corresponds to the maximum likelihood estimate [2]. If the noise on the measured signal is Gaussian (which is what is predicted by the majority of models [2,17]), the conditional probability $P(\mathbf{d}|\mathbf{a})$ is also Gaussian:

$$P(\mathbf{d}|\mathbf{a}) \propto \exp\left(-\frac{1}{2}(\mathbf{d} - \mathbf{R}\mathbf{Y}\mathbf{a})^T \mathbf{N}^{-1}(\mathbf{d} - \mathbf{R}\mathbf{Y}\mathbf{a})\right) \quad (2.9)$$

(where $\mathbf{N} = nn^T$ is the noise matrix where we have assumed each of the components is independent such as in Chapter 2.1) and:

$$P(\mathbf{a}|\{C_\ell\}) \propto \prod_{\ell=2}^{\ell_{max}} \left(\frac{1}{C_\ell}\right)^{\frac{2\ell+1}{2}} \exp\left(-\frac{2\ell+1}{2} \frac{\sigma_\ell}{C_\ell}\right) \quad (2.10)$$

(where $\sigma_\ell = \frac{1}{2\ell+1} \sum_m |a_{lm}|^2$ is the realization power spectrum) [11]. This posterior is then sampled in parameter space using a No-U-Turn sampler (Chapter 3.3). For this, the logarithm of the posterior is used:

$$\psi(\{C_\ell\}, \mathbf{a}) = \frac{1}{2}(\mathbf{d} - \mathbf{R}\mathbf{Y}\mathbf{a})^T \mathbf{N}^{-1}(\mathbf{d} - \mathbf{R}\mathbf{Y}\mathbf{a}) + \sum_{\ell} \left(\ell + \frac{1}{2}\right) \left(\ln(C_\ell) + \frac{\sigma_\ell}{C_\ell}\right) + D \quad (2.11)$$

(where $\psi(\{C_\ell\}, \mathbf{a}) = \log [P(\{C_\ell\}, \mathbf{a}|\mathbf{d})]$ and D is a constant which is differentiated to zero in the sampling).

To estimate the power spectrum, Planck uses a hybrid approach where the approach is split into a low and high ℓ regime. For the low ℓ , a quadratic maximum likelihood (QML) approach is used with a likelihood constructed by approximating the marginal distribution of the power spectrum derived from Gibbs sampling based component separation. The likelihood constructed in [2] presents a better approach to this regime since the Gaussian approach to $\{C_\ell\}$ breaks down at low ℓ . For the high ℓ , a pseudo- C_ℓ approach is used:

$$C_\ell = \frac{1}{2\ell+1} \sum_m \langle |a_{lm}|^2 \rangle \quad (2.12)$$

[55]. For this the Gaussian approach is a good approximation from the central limit theorem.

2.3 Sampling Algorithms

Hamiltonian Monte Carlo sampling is a Markov chain Monte Carlo sampling method which obtains a sequence of random samples from probability distribution functions. These don't require a full analytic description of the function for sampling to proceed - making them optimal for posterior functions in probabilistic inferences.

Hamiltonian Monte Carlo sampling is primarily used to tackle sampling in high parameter space. This uses gradient calculations to inform the sampling process and thus also its speed. It also uses a 'momentum', p , to aid the sampler such that it can be more efficient than a random walk. The 'Hamiltonian' reference comes from the involvement of a Hamiltonian term, H :

$$H = \sum_i \left(\frac{p_i^2}{2m_i} \right) + \psi(x) \quad (2.13)$$

and Hamilton's equations:

$$\frac{dx_i}{dt} = -\frac{\partial H}{\partial p_i} \quad (2.14)$$

and

$$\frac{dp_i}{dt} = -\frac{\partial H}{\partial x_i} = -\frac{\partial \psi(x)}{\partial x_i} \quad (2.15)$$

(where $\psi(x)$, is the negative log of the target distribution $P(x)$, m is the 'mass' involved and x are the parameters). An initial point in sample space is chosen and samples are drawn from a distribution proportional to the negative exponential of the Hamiltonian:

$$\exp(-H) = P(x) \prod_i \exp\left(-\frac{p_i^2}{2m_i}\right) \quad (2.16)$$

The momentum used for each step is drawn from a Gaussian distribution with variance m_i and zero mean. The system is then evolved for time, τ using Hamiltons equations. Then, a new point has been reached with a new Hamiltonian value, H' and this point is accepted with probability:

$$P_{acceptance} = \min(1, \exp(-\delta H)) \quad (2.17)$$

(where $\delta H = H' - H$) [2]. Therefore, if Hamiltons equations are integrated exactly, $\delta H = 0$ and thus the step is accepted. - this intergration is conducted using the leapfrog integration method (refer to Appendix B). Then, a new momentum is generated and the step is repeated. It is computationally expensive for to reject a lot of steps in high parameter space, but these equations allow for a high acceptance rate.

The log posterior function (2.11), can be sampled to infer the a_{lm} and the $\{C_\ell\}$. Since it has lots of input parameters, its samples move in high-parameter space. Since $\{C_\ell\}$ cannot be negative, it was found to be more computationally efficient to sample $\log(\{C_\ell\})$ instead of $\{C_\ell\}$. To keep a flat $\{C_\ell\}$ prior, an exponential prior on $\log(\{C_\ell\})$ was now used [2].

The caveat to HMC sampling is that it requires the sampler to specify at two parameters: a step size and the number of steps taken. The efficiency of the chain fringes on a good choice for this pair which can be computationally costly. Adaptive step size HMC is sometimes used to tackle the step size however, the number of steps still requires expensive runs [60].

The no U-turn sampler (NUTS), is an extension to HMC that eliminates the need to set a number of steps for each sample and instead picks an appropriate length to maximise the distance of each step and avoid random walk behaviour.

$$\frac{dQ}{dt} = (\theta - \theta^*)p \quad (2.18)$$

If we are moving from θ to θ^* , the algorithm runs leapfrog steps until θ^* starts to move back towards theta. At this point, (2.18) becomes less than zero and the sampler either accepts or rejects the sample using with probability $\exp(p)$.

This was the more optimal sampler, as you can see also from a pair of trace plots comparing them in Figure 2.1

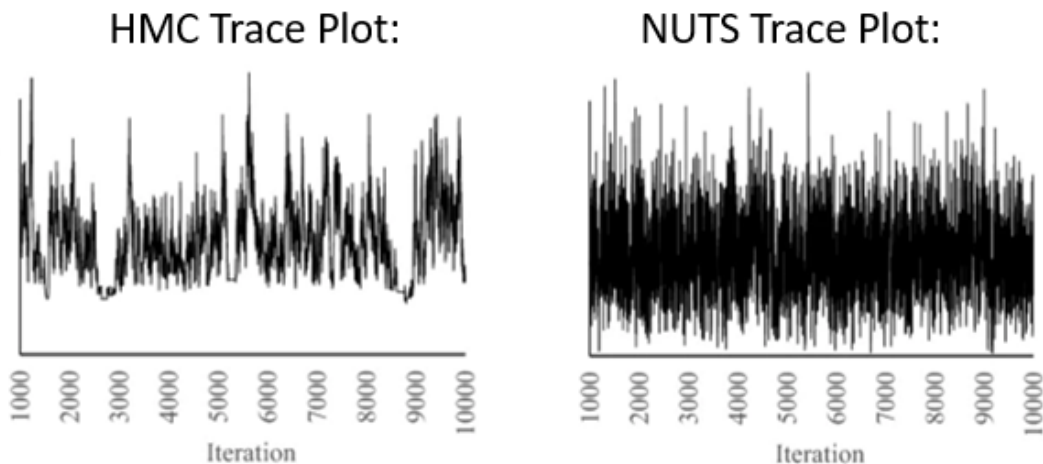


Figure 2.1: Comparison of the efficiency with which the NUTS and HMC sampling algorithms investigate a given sampling space. Adapted from [59].

2.4 Error Distributions

HMC has a large number of adjustable parameters, notably the masses. The distribution for the a_{lm} parameters is Gaussian and so we attempt to set the mass associated with each a_{lm} such that they are inversely proportional to the variance

of that a_{lm} .

The masses for the a_{lm} are estimated for a fixed power spectrum for which the variance is computed by:

$$\sigma_{a_{lm}}^2 = (C_\ell^{-1} + B_\ell N_{lm,l'm'}^{-1} \delta_{ll'} \delta_{mm'} B_\ell)^{-1} \quad (2.19)$$

$$\sigma_{C_\ell}^2 = \frac{2\ell + 1}{2f_{sky}} \left(C_\ell + \frac{N_\ell}{B_\ell^2} \right)^2 \quad (2.20)$$

(where f_{sky} is the fraction of the sky observed, N_ℓ are the elements of $\mathbf{N}$ and B_ℓ is the beam smoothing effect on the power spectrum) [2][47].

Chapter 3

Method

3.1 Sampling

When the log posterior (2.11) is calculated, the $\mathbf{Y}$ matrix becomes incredibly large as the sampling parameters, N_{side} and ℓ_{max} are increased. For double precision floating point accuracy, this matrix requires RAM:

$$M = 768l_{max}(l_{max} + 1)N_{SIDE}^2 \quad (3.1)$$

(where M is the RAM size needed to calculate the matrix of the spherical harmonics in bytes, for alms up to l mode l_{max} and for theta, phi for a map with pixels $= 12N_{SIDE}^2$). The full Planck data has $N_{SIDE} = 2048$ and $l_{max} = 2500$ which, using (3.1), requires 1831.8TB of RAM for the full set of spherical harmonics. For this project, the model was run on a machine with 25.5GB of RAM and so the sampling parameters had to be decreased from the Planck map to $N_{SIDE} = 128$ and $l_{max} = 60$.

There was another method to get around this RAM issue which only stored the legendre polynomials and then calculated the spherical harmonics for each pixel of BYa individually by multiplying by $e^{im\phi}$ on the fly. This reduced the size of the matrix stored by a factor of the number of unique θ in the RING scheme for a given N_{SIDE} . The problem with this however is that it increased the time taken for sampling by approximately the same amount. This method could be viable on a machine with a better processor, but for the purpose of this project, this was not explored further.

When sampling, the model takes the power spectrum, C_ℓ and a_{lm} as inputs and stores their values as it steps through parameter space. Using the realization power spectrum, σ_ℓ , the Blackwell-Rao estimator can be used to generate a smooth approximation to the marginal distributions:

$$Pr(\{C_\ell\}|\mathbf{d}) = \int Pr(\{C_\ell\}, \mathbf{a}|\mathbf{d})d\mathbf{a} = \int Pr(\{C_\ell\}|\mathbf{a})Pr(\mathbf{a}|\mathbf{d})d\mathbf{a} \quad (3.2)$$

which under the assumption that the CMB and by extension $\mathbf{a}$ is Gaussian,

$$Pr(\{C_\ell\}|\mathbf{d}) = \int Pr(\{C_\ell\}|\{\sigma_\ell\})Pr(\{\sigma_\ell\}|\mathbf{d})d\{\sigma_\ell\} \quad (3.3)$$

where

$$Pr(\{C_\ell\}|\{\sigma_\ell\}) = \prod_{\ell} \left(\frac{\sigma_\ell}{C_\ell} \right)^{\frac{2\ell-1}{2}} \exp \left(-\frac{2\ell+1}{2} \frac{\sigma_\ell}{C_\ell} \right) \quad (3.4)$$

For a given C_ℓ , the marginal distribution from M samples is therefore approximated by:

$$Pr(C_\ell|\mathbf{d}) \approx \frac{\sum_i Pr(C_\ell|\sigma_\ell^i)}{M} \quad (3.5)$$

where

$$Pr(C_\ell|\sigma_\ell) = \left(\frac{\sigma_\ell}{C_\ell} \right)^{\frac{2\ell-1}{2}} \exp \left(-\frac{2\ell+1}{2} \frac{\sigma_\ell}{C_\ell} \right) \quad (3.6)$$

3.2 Tensorflow

Tensorflow is a python package developed by Google which was crucial for this project, it has optimised sampling algorithms such as the NUTS which improved the efficiency of the model used.

Automatic differentiation is a group of computational techniques which allow the numerical calculation of the derivative of a function (automatic differentiation). In the model built, this was used for the calculation of Hamilton's equations. In automatic differentiation, the derivative is calculated for the HMC and NUTS algorithms, allowing for much faster sampling of the parameter space. The workings of automatic differentiation are described in [17].

3.3 Convergence

Once sampling the parameter space had been achieved, the convergence of the chain must be determined.

The first method of determining this is to observe how the log posterior changes as a function of step incidence. A 'burn-in' period should first be observed, followed by an exploration above and beyond some value.

The next test would be to plot trace plots of the parameters to determine if they demonstrated similar behaviour to the log posterior function (a burn-in phase followed by an approximately consistent sampling).

The final method to determine the convergence used the Gelman-Rubin statistic [43]. This value, R , determines the convergence of parameters by comparing the samples multiple chains. R is defined as:

$$R = \left(\frac{V}{W} \right)^{0.5} \quad (3.7)$$

where V is the 'pooled variance', and W :

$$W = \frac{1}{M} \sum_{m=1}^M \sigma_m^2 \quad (3.8)$$

(where σ_m^2 is the variance of the m^{th} chain, for m chains of N samples). An R value between 0.9 and 1.1 is typically said to be converged.

However, in certain circumstances, the Gelmann-Rubin statistic can have flaws [30]. It is therefore useful to estimate the number of independent samples have been taken in a chain of samples (the number of effective steps). Autocorrelation in a chain increases the uncertainty in the estimates that can be made of the number of effective samples in a chain. For a chain with no correlation, the number of effective samples is equal to the number of steps. If there is positive correlation, the number of effective steps is less than the number of steps and with negative correlation it is be more. It can be estimated using a Tensorflow function, `'mcmc.effective_sample_size'` [45].

Once the chain has been determined to have converged, the 'burn-in' stage needs be determined such that it can be discarded from the chain. This is because the mean parameters sampled are output of the model and so these samples where the chain has not yet burnt in taint the results. There are multiple methods to determine the sample at which the chain has be 'burnt-in' but that chosen was to determine the chain as 'converged' once the chain had reached the maximum value of the remainder of the chain.

Chapter 4

Results and Discussion

The log posterior function (2.11) was sampled using the NUTS with a power spectrum for $2 \leq \ell < 60$ and using a map pixelised at $N_{side} = 128$. 4 independent chains were sampled from the initial 'noisy map', each with 2400 samples. This sampling took 138 hours for the Google Colab server running with dual core processors, giving an a time per step of 52 seconds. This time however, does not consider the effect of take in to account the number of effective steps in the chain

4.1 Convergence

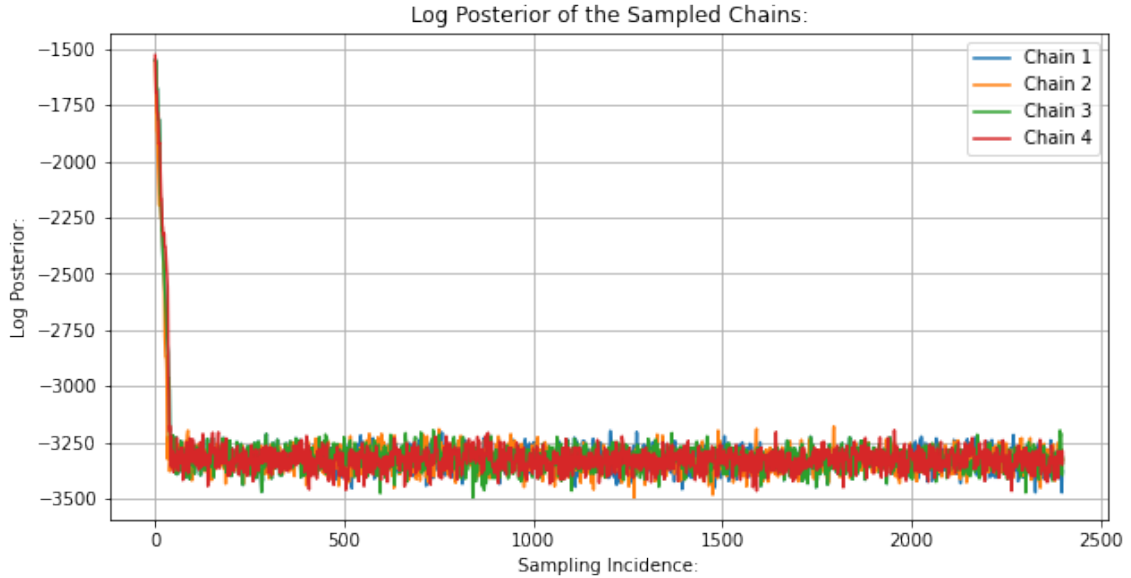


Figure 4.1: Sampling of the log posterior. This has the structure of an initial burn-in stage, followed by a consistent sampling demonstrating convergence.

Figure 4.1 clearly demonstrates this behaviour and so this passes the first convergence test. Following this, a burn-in value was selected as 42 steps, using the method described in Chapter 3.3.

The Gelman-Rubin statistic can then also be used to determine the convergence of results of the multiple chains. For the four chains sampled, a parameter is classified

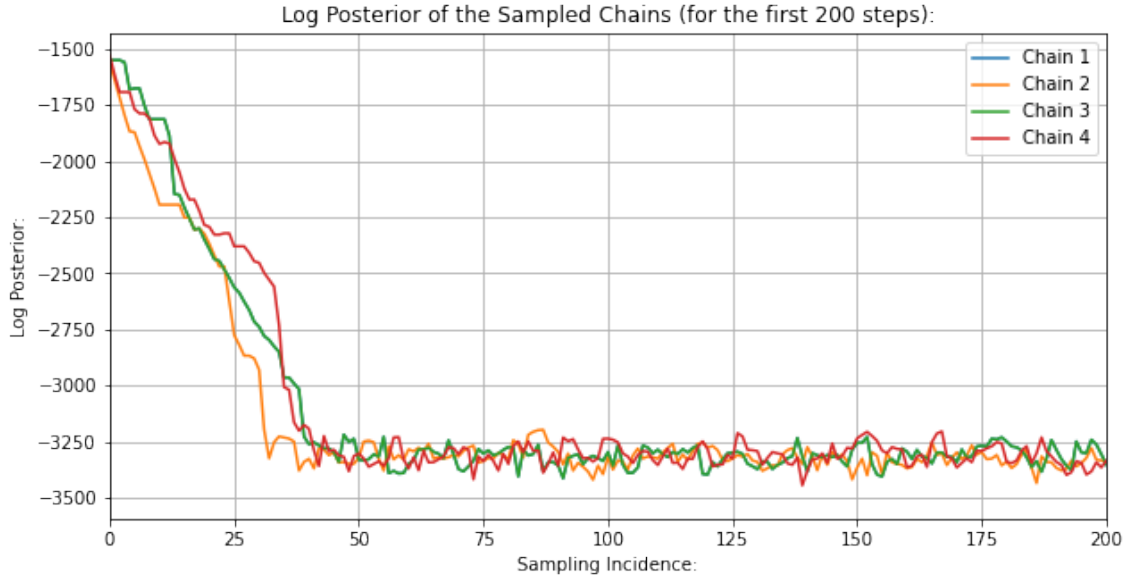


Figure 4.2: Sampling of the log posterior cut to the first 200 steps. This has the structure of an initial burn-in stage, followed by a consistent sampling demonstrating convergence and lead to a cut-off burn-in value to be chosen as 42.

as 'converged' if $0.9 \leq R \leq 1.1$. The values of R were calculated for the chains and plotted as histograms for Figure 4.5. Of the 3596 parameters used in the model - C_ℓ and a_{lm} , 6 C_ℓ and 0 a_{lm} had not converged. This gives success rate of convergence of 89.7% for C_ℓ and 100% for a_{lm} .

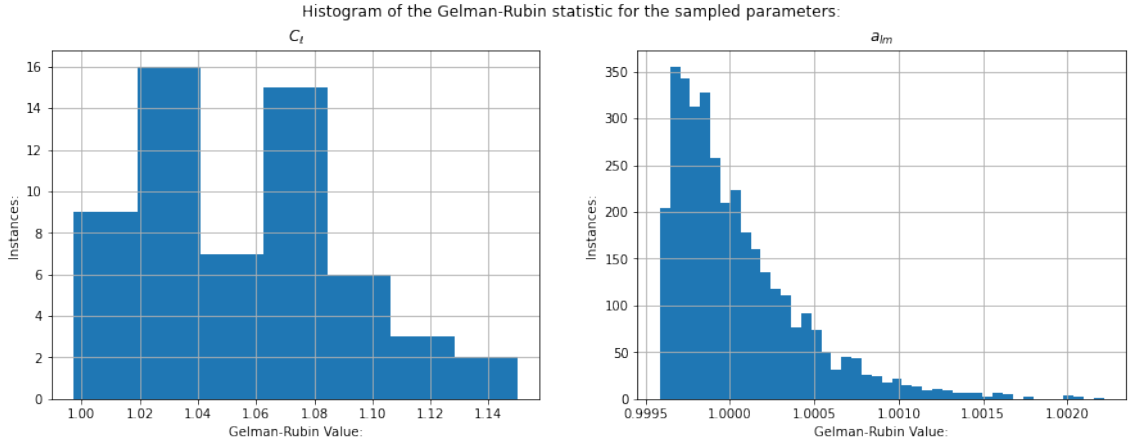


Figure 4.3: Histograms for the Gelman-Rubin statistic for each of the sampling parameters used in the model - C_ℓ (left) and a_{lm} (right). The number of bins used in each histogram was optimised to be 7 and 43 respectively.

As mentioned in section 3.4, the Gelman-Rubin statistic can in some cases be misleading for classifying convergence. The proposed alternative was to consider whether the chains sampled had a significant amount of correlation to determine if independent steps had been taken in the sample space. The amount of correlation in a chain can be determined by calculating the number of effective steps taken in parameter space, N_{eff} , and comparing it to the number of steps taken.:

The number of effective steps taken in each chain were marginally less than the

Number of Effective Steps, N_{eff}	
Chain 1	2297.2
Chain 2	2292.6
Chain 3	2270.5
Chain 4	2284.5

Table 4.1: The number of effective steps taken in parameter space, N_{eff} for each of the chains with the 'burn-in' samples removed. This is a lower bound estimate, calculated using the Tensorflow function, 'mcmc.effective_sample_size'. The number of effective samples taken in each chain was marginally less than the number of steps taken, 2358, indicating a minimal positive correlation in the chains.

number of steps taken. This indicates that there was positive correlation in the samples. However, the difference between the mean number of effective steps taken and the number of steps taken was very small when compared to the total number of steps taken and therefore it was concluded that there was insignificant correlation in the chains.

4.2 Timing

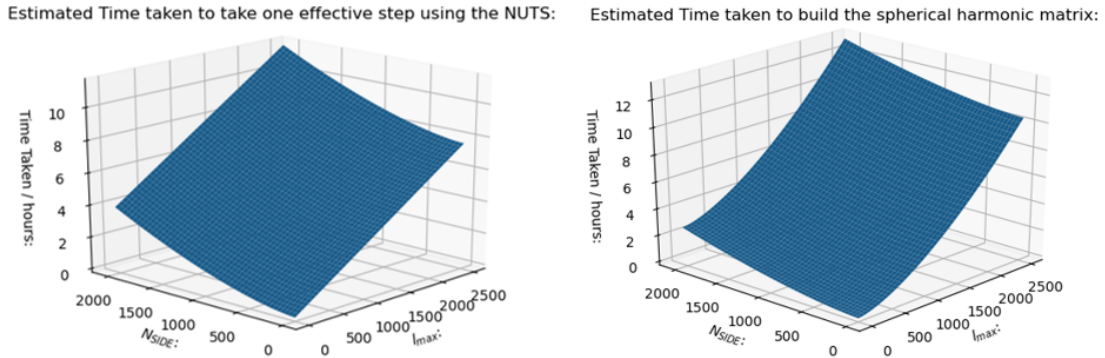


Figure 4.4: Extrapolated time taken to sample as a function of the two model parameters, by pre-loading the spherical harmonic matrix.

The time taken to sample as a function of the models parameters was extrapolated using the relationship that the time for the step as a function of the parameters scales as a 1.5 power law [2]. These times were considering the pre-loading the spherical harmonic matrix.

4.3 Power Spectrum

The mean sampled power spectrum was plotted as:

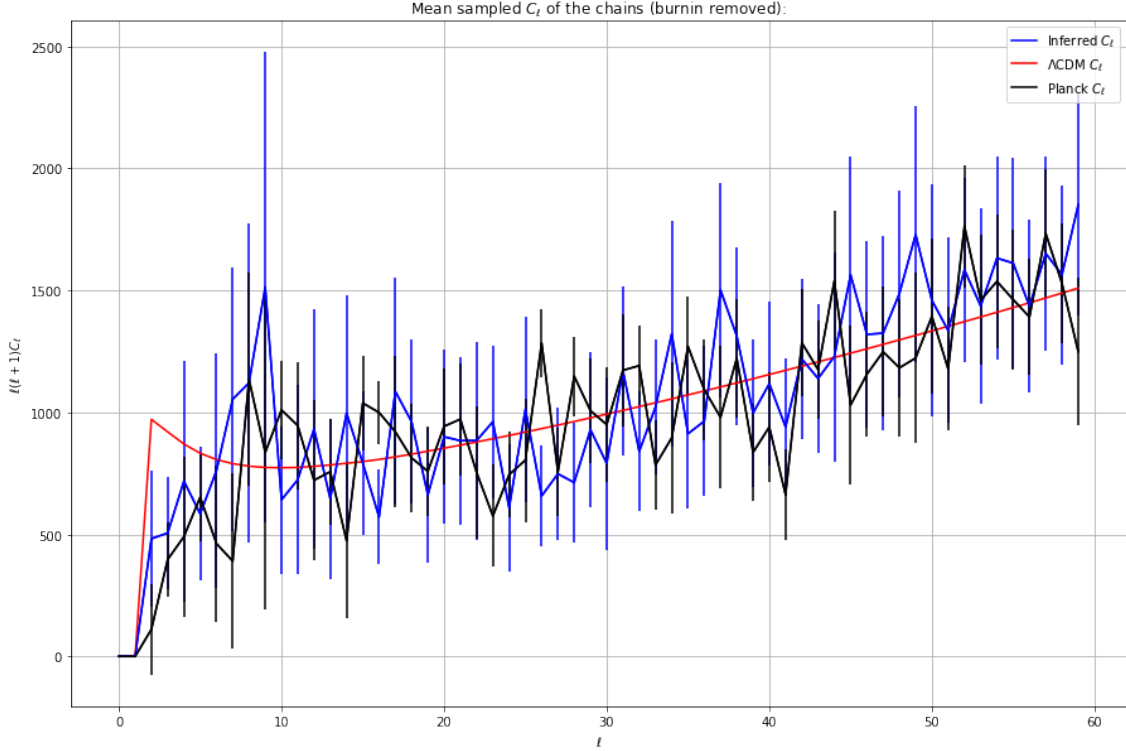


Figure 4.5: Power spectrum yielded by the Planck team through Gibbs sampling, compared to that of the Λ CDM model and the inferred power spectrum. This demonstrated an alignment between the spectra highlighting the efficacy of HMC and NUT samplers in this context.

This shows that the power spectrum sampled aligns with that of the Λ CDM model and so the efficient sampling will also be accurate at inferring cosmological parameters. This will be useful in the future as the resolution of CMB satellites only increases.

Chapter 5

Future outlook:

The first potential extension of the model would be the running of it on a machine with a greater processor and more RAM. This would lead to the ability of running the model with a greater N_{side} and l_{max} . The model could be extended to the full Planck data with a map pixelised at $N_{side} = 2048$ and multipoles $2 \leq \ell \leq 2500$, for which using (3.1), would require 1831.8TB of RAM for sampling with the pre-calculated spherical harmonics, or 149GB using the Legendre polynomials and calculating the on the fly (as in Chapter 3.1). The time taken per effective step extrapolated in Figure 4.4 will also improve since a processor with better capabilities will be able to sample through the parameter space faster.

The next logical step from an inferred power spectrum is the inference of cosmological parameters of the Universe. This is possible, due to the relationship between the CMB and the Universe in its formation. These parameters can be inferred using the structure of the TT power spectrum to determine these parameters to include regions of the Universe which we are not causally connected to today.

There are three methods of determining these parameters proposed by [35]:

- A direct non-linear fit based on the Levenberg-Marquardt algorithm (Press et al. 1992)
- Bayesian inference with Markov Chain Monte Carlo (MCMC)(Christensen et al. 2001)
- Peak-by-peak fits with a Gaussian fitting function

Of these methods, the MCMC methods seems most suitable to this report due to the experience with HMC sampling. This method analyses the characteristics of the peaks of the TT power spectrum to infer 6 parameters for a Λ CDM model - the Hubble constant, the Baryon density, the dark matter density, the curvature, the sum of the neutrino masses and the optical depth of the Universe [1].

With more time, the model can be used to collect more samples of the posterior distribution. This is not to the effect of further exploring the parameter space but instead to increase the accuracy of the Rao estimator [2]. At high ℓ especially, a very number of samples is needed to decrease this marginal distribution (3.6).

Chapter 6

Conclusion

A model was built to efficiently sample the posterior function of the CMB TT power spectrum. This sampling used the NUTS for its efficient sampling in high dimensional space and its ability to explore the non-gaussianity of the spectrum at low ℓ . The model was run on the Google Colab server which uses dual core processors and 25.5GB of RAM. For this space, four independent chains were sampled in a period of 138 hours (with an average time for an effective step of 54.3 seconds in the parameter space. This sampling was conducted with model input parameters, $N_{side} = 128$ and $l_{max} = 60$. The mean acceptance rate of the chains was 0.985 which indicates an efficient sampling of the posterior function - since it is computationally costly to reject samples in high parameter space.

The time taken to step through this parameter space could be optimised using a better machine to receive much smaller sampling times. The optimised sampling of the model compared to previous literature was attributed to the utilisation of Tensorflow and its optimised functions. The NUTS function they provide uses automatic differentiation to step through the sample space faster than other implementations of the NUTS.

The next extension of the project, once the model has been used on a better machine would be to infer the cosmological parameters which compose the Universe and compare them to those of the Λ CDM model.

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Appendix A

CMB Temperature Map Measurements

When a satellite records microwave radiation across the sky, there are contaminations from external galactic sources to measure the CMB. These are outlined in Figure A.1:

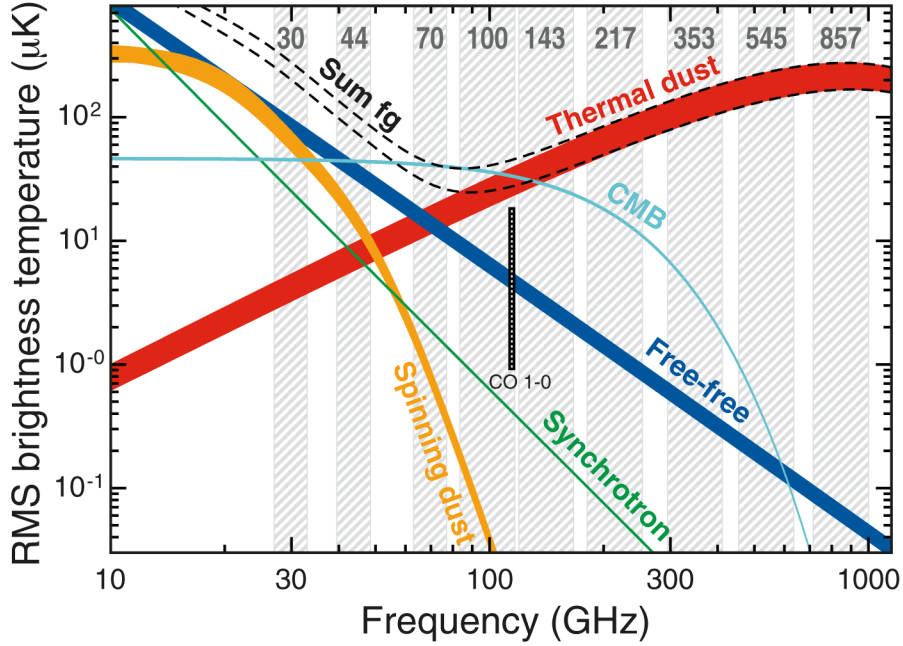


Figure A.1: The brightness of sources of radiation in the milky way as a function of frequency. There is a minimum in the interfering signals between 70-80GHz where the CMB signal is greater than the total galactic signal (Sum fg). The nine different frequency bands that Planck simultaneously records in are outlined in grey [46].

The synchrotron emission dominates this foreground at frequencies ≤ 30 GHz whilst thermal dust takes over at frequencies > 70 GHz. There is a minimum in this interference distribution between 70-80GHz where the CMB is the stronger signal. To distinguish the foreground from the CMB, a process called 'component separation' is conducted on the recorded maps. There are different methods to conduct this process, but Planck uses five: 'Commander', 'NILC', 'SEVEM', 'SMICA' and 'GNILC'. These utilise maps recorded in 9 different frequency bands to classify the

galactic foreground [54].

This process also uses masks that remove regions of the sky that are obstructed; for example, the Planck team used a mask called the 'U73' - where 23% of the sky was removed for their final CMB TT map [50].

Appendix B

Leapfrog Method

The leapfrog method is used for the numerical integration of differential equations. This occurs for a varied time, τ :

$$\tau = N\epsilon \quad (\text{B.1})$$

(where N is the number of steps taken and ϵ is the total number of steps taken).

The evolution of time of the Hamiltonian dynamics of the HMC sampler were simulated using the “leapfrog” integrator, which proceeds according to the updates:

$$r^{t+\frac{\epsilon}{2}} = r^t + (\epsilon/2)\nabla_{\theta}\mathcal{L}(\theta^t) \quad (\text{B.2})$$

$$\theta^{t+\epsilon} = \theta^t + \epsilon r^{t+\epsilon/2} \quad (\text{B.3})$$

$$r^{t+\epsilon} = r^{t+\epsilon/2} + (\epsilon/2)\nabla_{\theta}\mathcal{L}(\theta^{t+\epsilon}) \quad (\text{B.4})$$

(where r^t and θ^t denote the values of the momentum and position variables r and θ at time t and ∇_{θ} denotes the gradient with respect to θ). Since the update for each coordinate depends only on the other coordinates, the leapfrog updates are volume preserving. This means that the volume of a region remains unchanged after mapping each point in that region to a new point via the leapfrog integrator.

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